Question Bank

Course Code: MAT3003

Course Title: Probability, Statistics & Reliability

Module-1: Probability Theory

1. Seventy percent of the employees of the XYZ corporation are college graduates. Of these, ten percent are in sales. Of the employees who did not graduate from college, eighty percent are in sales. What is the probability that:
    - An employee selected at random is in sales?
    - An employee selected at random is neither in sales nor a college graduate?
2. In an assembly-line operation, 1/3 of the items being produced are defective. If three items are picked at random and tested, what is the probability that at least one of them will be defective?
3. A problem in statistics is given to three students A, B, and C, whose chances of solving it are 1/2, 1/3, and 1/4, respectively. What is the probability that the problem is solved?
4. The odds against A solving a certain problem are 8 to 6, and the odds in favor of B solving the same problem are 14 to 10. What is the probability that if both of them try, the problem would be solved?
5. In a bolt factory, machines A, B, and C manufacture 25%, 35%, and 40% of the total output, respectively. Out of their output, 5%, 4%, and 2% are defective bolts. A bolt is drawn from the production and is found defective. What are the probabilities that it was manufactured by A, B, and C?
6. From a city population, the probability of selecting:

    (i) A male or a smoker is 7/10

    (ii) A male smoker is 2/5

    (iii) A male, if a smoker is already selected, is 2/3
    Find the probability of selecting:

        (a) A non-smoker

        (b) A male

        (c) A smoker, if a male is first selected
7. A and B toss a coin alternately on the understanding that the first to obtain heads wins the toss. Show that their respective chances of winning are 2/3 and 1/3.
8. Two cards are drawn successively and without replacement from an ordinary deck of playing cards. Compute the probability of drawing:
    - Two hearts

- A heart on the first draw and a club on the second draw
- A heart on the first draw and an ace on the second draw

9. Let A and B be independent events with P(A) = 1/4 and P(B) = 2/3. Compute:

   (a) $P(A \cap B)$

   (b) $P(A \cap B')$

   (c) $P(A' \cap B')$

   (d) $P[(A \cup B)']$

   (e) $P(A' \cap B)$

10. The probability that a regularly scheduled flight departs on time is P(D) = 0.83; the probability that it arrives on time is P(A) = 0.82; and the probability that it departs and arrives on time is $P(D \cap A) = 0.78$. Find the probability that a plane:

    (a) Arrives on time given that it departed on time

    (b) Departed on time given that it has arrived on time

11. Suppose that we have a fuse box containing 20 fuses, of which 5 are defective. If 2 fuses are selected at random and removed from the box in succession without replacing the first, what is the probability that both fuses are defective?

12. A red die and a white die are rolled. Let event C = {5 on red die} and event D = {sum of the dice is 11}. Of these 36 equally likely outcomes:

    - 6 are favorable to C
    - 2 are favorable to D
    - 1 is favorable to $C \cap D$
      Hence, show that C and D are independent.

13. A class in advanced physics is comprised of 10 juniors, 30 seniors, and 10 graduate students. The final grades show that 3 of the juniors, 10 of the seniors, and 5 of the graduate students received an A for the course. If a student is chosen at random from this class and is found to have earned an A, what is the probability that the student is:

    - A junior
    - A senior
    - A graduate student

14. In a certain factory, machines I, II, and III produce springs of the same length. Their respective defect rates are 2%, 1%, and 3%. The proportion of total production by each machine is 35%, 25%, and 40%, respectively. If one spring is selected at random, find the probability that it is defective.

15. At a hospital's emergency room, patients are classified as:

- 20% critical (30% mortality rate)
- 30% serious (10% mortality rate)
- 50% stable (1% mortality rate)
  Given that a patient dies, what is the conditional probability that the patient was classified as critical?

16. Two processes of a company produce rolls of materials:
    - Process I: 3% defective, produces 60% of output
    - Process II: 1% defective, produces 40% of output
      Given that a roll is defective, what is the conditional probability that it is from Process I?
17. A rare disease exists in which only 1 in 500 are affected. A test for the disease gives:
    - 95% accuracy for a correct positive result
    - 1% false positive rate
      If a randomly selected individual tests positive, what is the probability that they actually have the disease?
18. Tickets numbered 1 to 20 are mixed up, and then a ticket is drawn at random. What is the probability that the ticket drawn has a number which is a multiple of 3 or 5?
19. A man and his wife appear for an interview for two vacancies in the same post. The probability of the husband's selection is 1/7, and the probability of the wife's selection is 1/5. What is the probability that only one of them is selected?
20. A professor is creating an exam with eleven questions from a test bank of 15 questions. In how many ways can he select and arrange the questions?
21. Let A and B be two events of a random experiment. The probability that at least one of the two events occurs is 0.6. The probabilities of event A occurring and event B occurring are 0.3 and 0.4, respectively. Find the probability that event A occurs given that event B has occurred.
22. Students from three colleges X, Y, and Z have participated in a competition:
    - 30% from college X, 20% from college Y, 50% from college Z
    - 50% of X, 40% of Y, and 60% of Z participants are girls
      If a girl is randomly selected, what is the probability that she is from college X? (Answer correct to three decimal places.)